

BuktiSaham

Product Requirements + Technical Requirements Implementation Blueprint for an Explainable Indonesian Equity Research Platform

Working name • Implementation-aligned design • Version 1.1 • 19 September 2026

Reader contract

Product and governance sections define what may ship. Data, quant and engineering sections define deterministic behavior. Security, SLO and operational sections define when the platform may remain available. Once this document is approved, it can be used as the baseline for repository scaffolding, database migrations, API contracts, tests and implementation tickets.



Figure 1. North-star product loop: evidence precedes analysis, publication is gated, and every view is scored after issuance.

Document basis: this version uses the uploaded implementation blueprint as a reference, retains its evidence-first and point-in-time principles, and makes the product behavior, data contracts, module boundaries, APIs, schemas, testing and delivery plan more implementation-ready.

Document control and how to use this blueprint

Field	Value
Status	Proposed for product, engineering, quant, data, security and compliance review
Intended next step	Approve / amend decisions → freeze v1.0 baseline → create implementation epics and repository scaffold
Primary journey	Create Investment Task → Run Analysis → Recommendation with Evidence → Save Decision Packet → Monitor Outcome
MVP audience	Analyst-supervised research and paper decisions; no autonomous personalized advice or broker execution
Architecture posture	Contract-enforced modular monolith + isolated ingestion/compute workers; extraction to services only when justified
Core invariant	A published recommendation is immutable, reproducible from versioned inputs, and auditable to evidence

Contents

1. Executive summary and approval decisions
 2. Product concept, users and boundaries
 3. End-to-end journeys and UX requirements
 4. Functional requirements and acceptance criteria
 5. Recommendation, evidence and decision methodology
 6. Indonesian-market calculations, valuation and simulation
 7. Data sources, point-in-time pipeline and data contracts
 8. Target architecture and component design
 9. Domain model, database schema and data retention
 10. API, event and workflow contracts
 11. AI usage and guardrails
 12. Security, compliance and entitlement controls
 13. Non-functional requirements and SLOs
 14. Testing and quality strategy
 15. Deployment, observability and disaster recovery
 16. Delivery roadmap, backlog and ownership
 17. Approval checklist and handoff to implementation
- Appendix A. API examples
- Appendix B. Database table blueprint
- Appendix C. Reference and verification register

Decision labels used in this document

MUST = release-blocking requirement. SHOULD = expected unless an approved ADR explains otherwise. MAY = optional extension. “Proposed” means a design choice in this blueprint; “verified source fact” means it is grounded in an official or referenced external source and still requires effective-date handling in production.

1. Executive summary and approval decisions

BuktiSaham is an evidence-first research workspace for Indonesian listed equities. It is intentionally not a “black-box stock picker.” Its product value is that a user can create a research task, run reproducible analysis, receive a scenario-led research view, inspect every material source and assumption, and later see exactly how that issued view performed after costs and corporate actions.



Figure 2. The same user journey is also the governance boundary: no generator can publish directly.

1.1 Decisions to approve now

ID	Decision	Why it matters	Reversible?
D-01	Use “BuktiSaham” as the working name; legal/trademark review before external use.	Evidence-led promise; locally understandable; avoids certainty language.	Yes
D-02	MVP supports research + paper decisions; no personalized suitability, custody, leverage or executable orders.	Reduces regulatory and operational scope while product-market fit is tested.	Yes
D-03	Deterministic calculations are authoritative; local Ollama may summarize or challenge only.	Numeric results remain reproducible and testable.	Mostly
D-04	All market rules, fees, taxes, entitlements and calendars are effective-dated data, never hard-coded constants.	IDX rules and investor/broker treatment change over time.	No for data model
D-05	Published recommendation versions are append-only; revisions create a new version.	Preserves auditability and honest outcome scoring.	No
D-06	Start with a modular monolith + isolated workers + PostgreSQL/Redis + local Ollama.	Fast enough for MVP without losing domain boundaries.	Yes
D-07	Use point-in-time/bitemporal semantics for security master, filings, prices, rules and corporate actions.	Prevents look-ahead bias and makes historical replay defensible.	No
D-08	Publication is fail-closed when evidence, entitlement, policy, reviewer or content hash is missing.	Protects trust and data rights.	No

1.2 Product success definition

A successful release lets a user answer five questions without trusting a black box:

1) What was observed? 2) What was calculated? 3) Which assumptions drive the result? 4) What would invalidate the view? 5) How did the full population of issued views perform after realistic costs?

Metric	Initial target	Release use
Evidence completeness	100% of mandatory claims trace to immutable evidence	Hard publish gate
Deterministic reproducibility	100% sampled reruns with same snapshots/rule versions reproduce outputs	Hard beta gate
Scenario comprehension	≥90% of test users identify horizon, downside and invalidation	UX gate
Prohibited-claim escape	0 in release policy test suite	Kill switch / release gate
Corporate-action reconciliation	100% material events reconciled before affected universe is actionable	Universe gate
Run completion	≥99% non-data-failure runs finish within target window	Operational gate

2. Product concept, users and boundaries

2.1 Product promise

“See the evidence behind every market view.” The platform converts official or licensed Indonesian market data into versioned decision packets containing a research view, scenario ranges, valuation, risk, execution feasibility, catalysts, invalidation logic, evidence lineage, reviewer status and eventual outcome score.

2.2 Target users and jobs-to-be-done

User	Primary job	Must receive	MVP boundary
Retail research reader	Understand a stock thesis and downside without reading every raw filing.	Plain-language scenario, source links, freshness, risks, invalidation.	No personalized suitability claim.
Professional user	Compare candidates and stress assumptions.	Exportable inputs, sensitivity, liquidity, method breakdown, scorecards.	Entitlement-controlled data.
Analyst	Author and revise defensible views.	PIT evidence, sector model, run diff, review workflow.	Cannot self-approve.
Compliance reviewer	Prevent prohibited, misleading or unlicensed publication.	Policy result, conflicts, audience rights, immutable audit package.	Independent approval.
Data / quant steward	Reproduce calculations and monitor quality.	Versioned lineage, backtests, drift, reconciliation.	No silent data repair.
Platform operator	Keep the service safe and recoverable.	SLOs, runbooks, kill switches, audit visibility.	Cannot override governance without break-glass logging.

2.3 Scope by release tier

Capability	P0 MVP	P1	P2 / gated
Task setup + analysis run	Yes	-	-
Evidence-linked recommendation	Yes	-	-
Analyst review + immutable versioning	Yes	-	-
Sector-specific valuation	Selected sectors	Expand coverage	Advanced specialty models
Portfolio context	Basic feasibility	Concentration + scenario loss	Optimization
Alerts	Stale/invalidation only	Catalyst & material-change diff	Personalized alerts only after legal review
Mobile	Responsive web	Native shell optional	-
Broker handoff	No	No	Non-executable intent only after separate legal/broker approval
Order routing / custody	No	No	Out of scope unless a future separately approved program

2.4 Explicit non-goals

- Guaranteeing market direction or return.
- Autonomous AI deciding BUY/SELL without deterministic policy and human governance.
- High-frequency or intraday market-making signals.
- Using today's constituent list to backtest historical recommendations.
- Rewriting old recommendation versions after a later data correction.
- Treating unlicensed public pages as permission for bulk production reuse.

3. End-to-end journeys and UX requirements

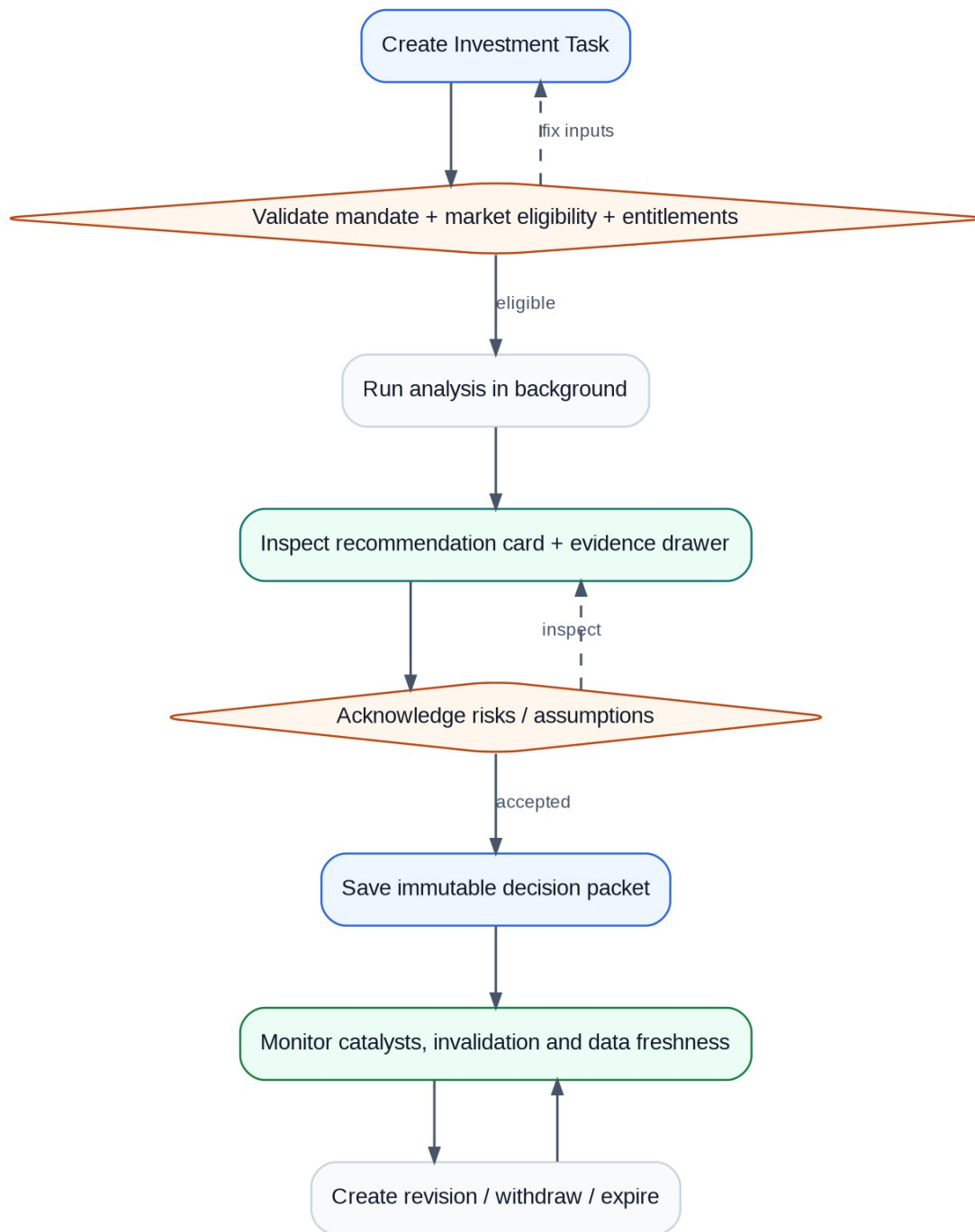


Figure 3. Primary user journey. A user can explore, but cannot save an actionable packet until validation and risk acknowledgement are complete.

3.1 Create Investment Task

The task is the reproducibility anchor. It captures what the user is asking, the eligible universe, investment horizon, capital assumptions, data entitlements and exact “as-of” time. Changing any of these creates a new task version or rerun, not an invisible mutation.

Field group	Required fields	Validation
Identity	task_id, tenant_id, creator_id, title, created_at	Server-generated IDs; actor context required.
Research intent	objective, universe, benchmark, horizon, mandate	Unknown universe or unsupported mandate blocks run.

Field group	Required fields	Validation
Capital/execution	capital_idr, max_position_pct, liquidity_limit, cost_profile	Decimal values only; positive capital; profile must be effective at as_of.
Time	as_of, market_timezone, data_delay_mode	Asia/Jakarta interpretation; UTC storage.
Governance	audience, entitlement_profile, disclaimer_profile	Missing entitlement/disclaimer profile blocks actionable output.

3.2 Run Analysis

Runs are asynchronous and idempotent. The UI immediately returns a run_id, then streams or polls stage status. Each stage records start/end time, input snapshot hashes, output hash and error category. A rerun with the same task version and same idempotency key must not duplicate the run.

Stage	What happens	Failure behavior
1. Universe freeze	Resolve effective security master, board, status, sector and eligibility.	Block securities with unresolved master conflicts.
2. Evidence snapshot	Seal prices, filings, corporate actions, macro and rules known at as_of.	Stale/missing mandatory source → preview-only or blocked.
3. Normalize	Units, currency, accounting periods, corporate-action adjustments.	Quarantine affected issuer; preserve raw input.
4. Features	Compute fundamental, technical, liquidity and event features.	Partial feature set allowed only if policy defines applicability.
5. Valuation	Run sector-appropriate methods + sensitivities.	Method failure is explicit; never silently replace with an invalid method.
6. Scenarios	Generate bull/base/bear ranges and probabilities from deterministic inputs.	Probabilities must sum to 1 within tolerance.
7. Decision metrics	Compute expected net return, downside, confidence and execution feasibility.	Calculation error blocks recommendation.
8. Evidence binding	Map claims to evidence IDs and transform lineage.	Any material unbound claim blocks publish.
9. Policy evaluation	Rights, freshness, audience, conflicts, thresholds, approval requirement.	Fail closed for actionable publication.
10. Draft packet	Create draft recommendation version and explain deltas.	AI narrative may fail independently; numeric packet remains available.

3.3 Recommendation card

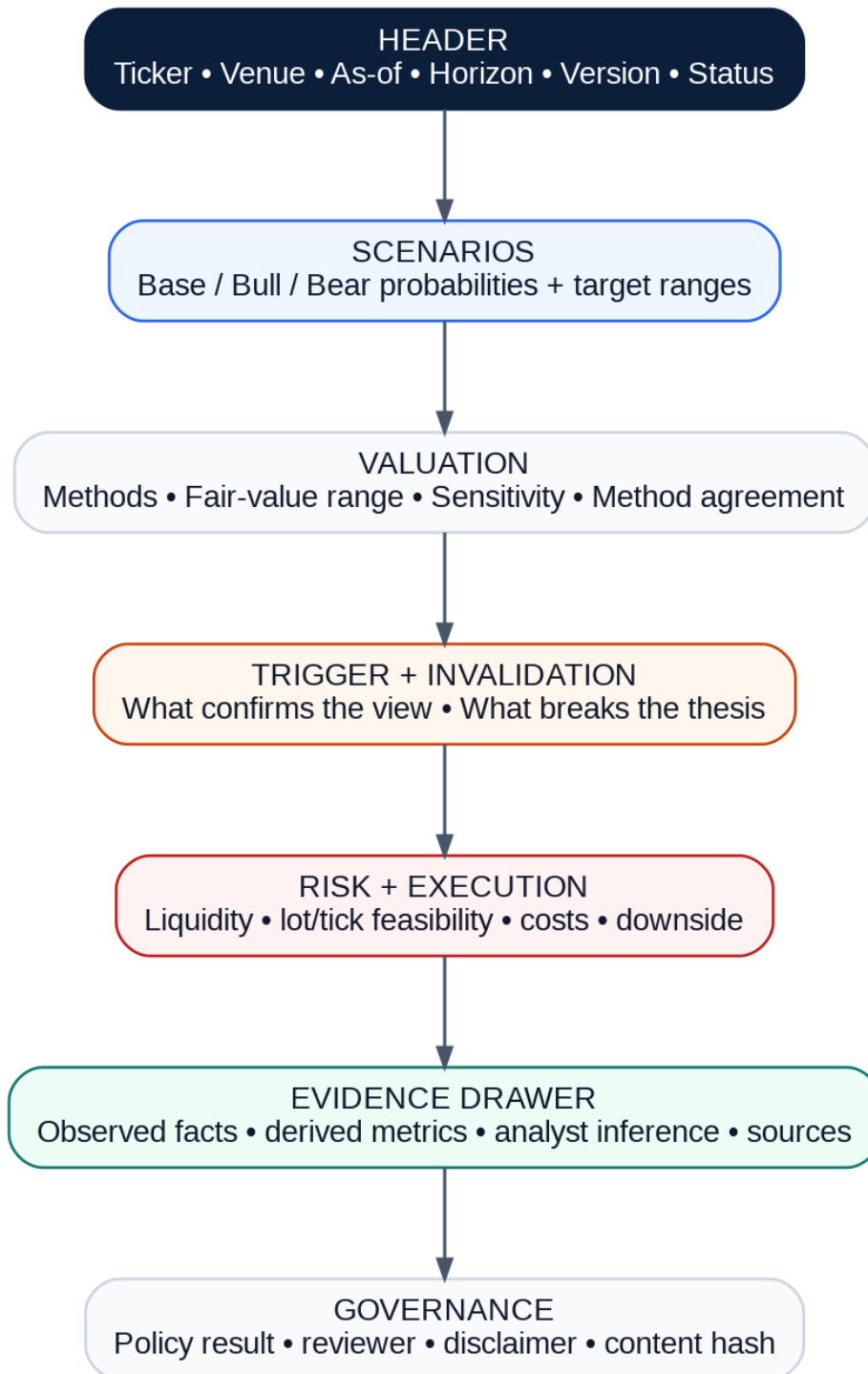


Figure 4. Recommendation card information architecture. The evidence drawer is a first-class product surface, not a footnote.

3.4 Recommendation lifecycle

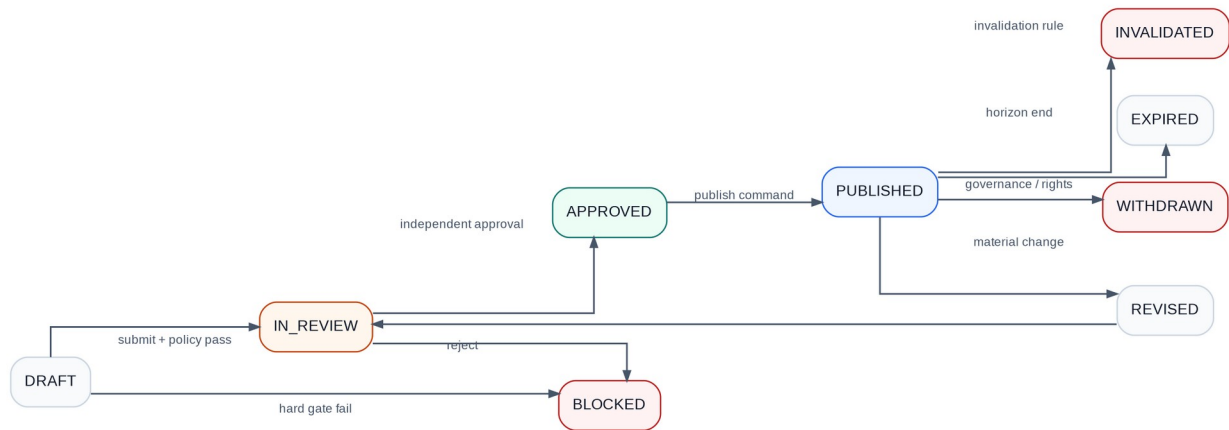


Figure 5. Append-only recommendation lifecycle. Published history is never overwritten.

4. Functional requirements and acceptance criteria

ID	Pri	Module	Normative requirement	Acceptance criterion
FR-01	P0	Task setup	System MUST capture universe, horizon, mandate, capital, cost/tax profile, entitlement and as-of time.	Invalid mandatory inputs block run with field-level reason.
FR-02	P0	As-of reproducibility	Every run MUST pin all source, rule, model and parameter versions.	A rerun from saved packet reproduces deterministic outputs within decimal tolerance.
FR-03	P0	Universe freeze	System MUST use effective-dated security master; historical runs may not use today's list.	Fixture with delisted/suspended names reproduces historical eligibility.
FR-04	P0	Evidence drawer	Every material claim MUST resolve to source snapshot, observed/effective times, transform lineage and content hash.	100% of publishable material claims pass referential-integrity checks.
FR-05	P0	Statement labels	UI MUST distinguish observed fact, derived indicator, analyst inference and recommendation.	≥90% classification accuracy in usability test.
FR-06	P0	Financial normalization	System MUST preserve reported and normalized values with mapping version.	Golden filings reconcile to source totals and period logic.
FR-07	P0	Corporate actions	Splits, dividends, rights and symbol changes MUST be event-sourced and reconciled.	Adjusted share/price/cash path balances for 100% material events.
FR-08	P0	Valuation routing	System MUST select only approved sector methods and declare unavailable methods.	Bank fixture never uses EV/EBITDA as primary; telecom lease basis is consistent.
FR-09	P0	Scenario engine	System MUST output base/bull/bear probabilities and ranges from versioned assumptions.	Probability sum = 1.000 ±0.001; assumption diff is machine-readable.
FR-10	P0	Local execution	Lots, ticks, calendars and costs MUST use effective-dated rule tables.	Reference fixtures match then-effective IDX rules.
FR-11	P0	Recommendation policy	Action/status MUST be a deterministic function of eligibility, evidence, return/downside, confidence and execution constraints.	Policy test matrix covers all action/abstain branches.
FR-12	P0	Review workflow	Author, reviewer and publisher duties MUST be separable.	Self-approval and unauthorized transitions are rejected.
FR-13	P0	Immutable versions	Published recommendation versions MUST be append-only.	No UPDATE path exists for sealed numeric/evidence payload; revision uses supersedes_id.

ID	Pri	Module	Normative requirement	Acceptance criterion
FR-14	P0	Outcome ledger	Issued/revised/invalidated/expired/withdrawn versions MUST remain in complete aggregate scoring.	Ledger count reconciles exactly to publication events.
FR-15	P0	Policy fail-closed	Missing mandatory policy dependency MUST block publication.	Chaos test of policy service does not leak actionable content.
FR-16	P0	Audit export	Authorized reviewer MUST obtain a signed package of packet, evidence index, policy and approvals.	Export hash verifies; unauthorized actor gets 403.
FR-17	P1	Portfolio context	System SHOULD calculate lot feasibility, concentration, liquidity days and scenario loss.	Reference portfolio fixtures match approved formulas.
FR-18	P1	Alerts	System SHOULD explain exact evidence/assumption delta causing stale, revision or invalidation.	Alert deep-links to version diff and triggering evidence.
FR-19	P1	Comparison	User SHOULD compare candidates on normalized valuation, evidence completeness, downside and liquidity.	Same as-of and cost profile enforced for comparison.
FR-20	P1	Exports	User SHOULD export permitted derived tables with entitlement watermark and provenance.	Restricted datasets never appear in export payload.
FR-21	P2	Broker handoff	System MAY export non-executable order intent only after separate approvals.	Feature flag defaults off; no live execution route in MVP.

5. Recommendation, evidence and decision methodology

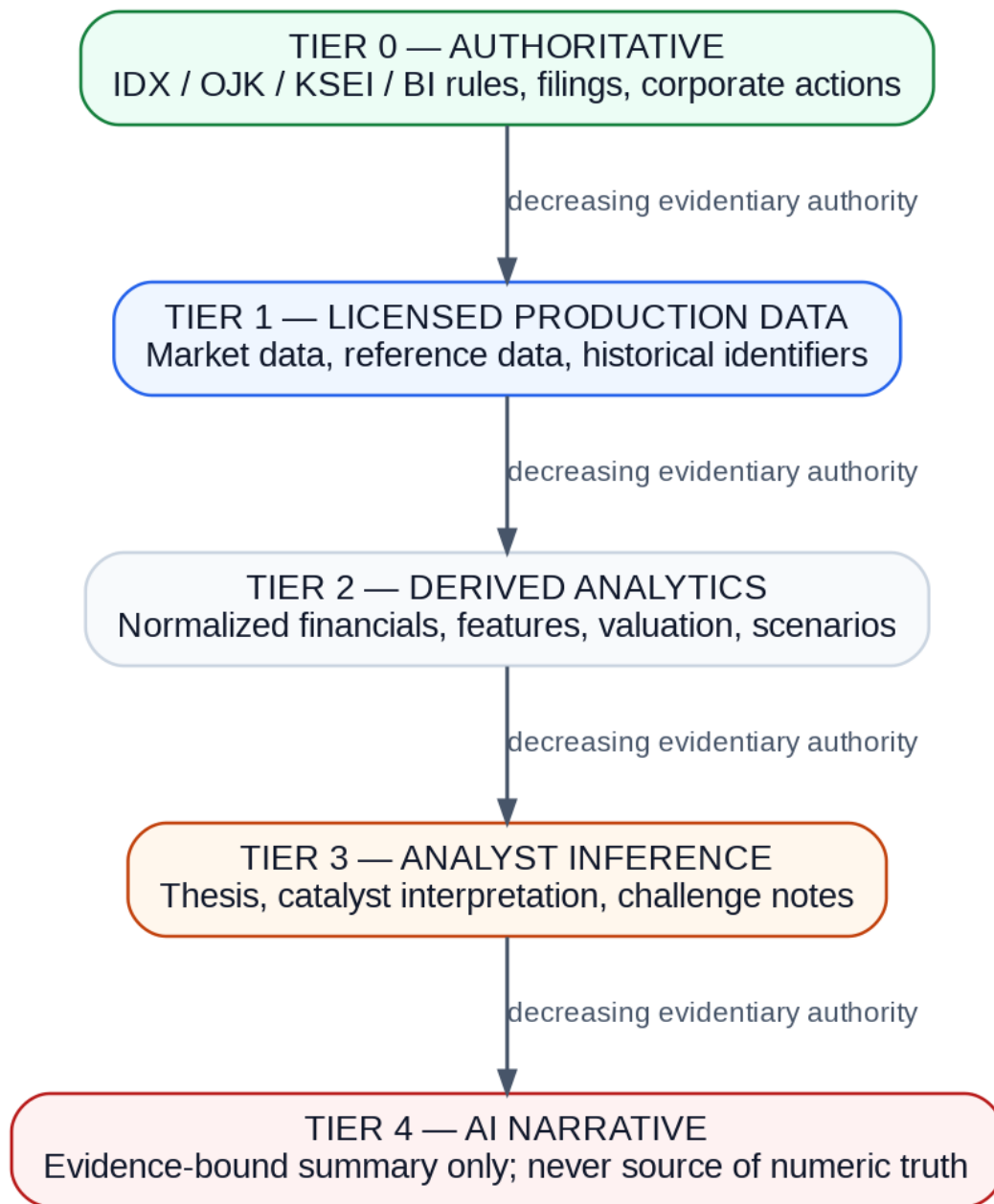


Figure 6. Evidence hierarchy. Visual emphasis never upgrades a weak evidentiary class.

5.1 Statement taxonomy

Label	Definition	Example	May change numeric result?
Observed data	Direct observation from approved source snapshot.	FY2025 revenue reported in issuer filing.	Source correction can change downstream results.
Derived indicator	Deterministic calculation over observed data.	ROE, EBITDA margin, 20-day volatility.	Yes, via deterministic formula.
Analyst inference	Human interpretation of evidence.	Margin pressure appears driven by imported inputs.	Cannot directly override deterministic numbers.
Recommendation	Policy outcome over scenario/risk/evidence metrics.	Research view: BUY / WATCH / HOLD / REDUCE / AVOID / NO-RATING.	Only via versioned policy inputs.

Label	Definition	Example	May change numeric result?
AI narrative	Generated explanation of already-approved evidence and outputs.	Summary of why base-case fair value changed.	No. Numeric/action mutation forbidden.

5.2 Scoring framework

The scoring system intentionally separates “how attractive the company looks” from “how safe it is to issue a view.” Missing information should reduce evidence completeness, not automatically make a company fundamentally bad.

Proposed score decomposition

Fundamental Attractiveness F (0–100) = weighted applicable rule results. Technical Context T (0–100) = trend, momentum, regime, volume and structure. Evidence Completeness E (0–100) = supported applicable evidence / required applicable evidence. Confidence C (0–100) = data quality + valuation agreement + forecast stability + liquidity/execution + catalyst evidence. Expected Net Return ERnet = probability-weighted scenario return including dividends minus effective transaction/tax/impact costs.

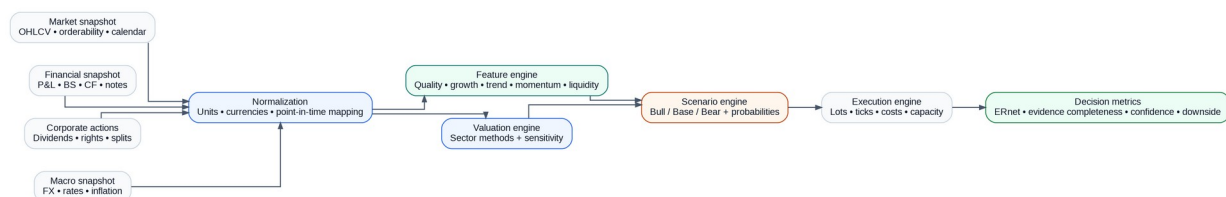


Figure 7. Deterministic research engine. The recommendation policy consumes outputs; it is not embedded inside valuation or AI code.

5.3 Proposed policy logic

Exact thresholds are configuration, not code constants. Product, quant and compliance must approve the first production parameter set after shadow calibration. The following decision order is normative even if thresholds change:

1. Check legal/audience/data entitlement eligibility. If false → NO-RATING / BLOCKED.
2. Check mandatory data freshness, corporate-action reconciliation and evidence completeness. If below hard minimum → NO-RATING or preview-only.
3. Check executability: tradable status, lot/tick feasibility, liquidity/capacity and cost model. If materially non-executable → AVOID or NO-RATING depending policy.
4. Compute fair-value distribution and scenario expected net return after effective costs.
5. Apply downside guardrail and confidence floor. A positive expected return cannot override excessive bear-case downside or weak evidence.
6. Require method agreement when multiple methods are expected for the sector; explicit single-method exceptions need an approved reason code.
7. Generate research action/status and reason codes. Human review may reject or request revision but must not silently edit deterministic outputs.

```
def evaluate_research_view(m):
    if not m.policy_eligible:
        return View.NO_RATING, ["POLICY_INELIGIBLE"]
    if m.evidence_completeness < cfg.min_evidence or m.data_stale:
        return View.NO_RATING, ["INSUFFICIENT_EVIDENCE"]
    if not m.executable:
        return View.AVOID, ["EXECUTION_CONSTRAINT"]
    if m.bear_downside > cfg.max_bear_downside:
        return View.WATCH, ["DOWNSIDE_GUARDRAIL"]
    if m.confidence < cfg.min_confidence:
        return View.WATCH, ["LOW_CONFIDENCE"]
    if m.expected_net_return >= cfg.buy_hurdle and m.valuation_methods_agree:
        return View.BUY, ["RETURN_HURDLE_MET"]
    if m.expected_net_return <= cfg.reduce_hurdle:
        return View.REDUCE, ["NEGATIVE_EXPECTED_RETURN"]
    return View.HOLD, ["NEUTRAL_BAND"]
```

5.4 Astronacci reference: what to borrow, what not to copy

Borrow as presentation principle	Do not depend on
Scenario-led presentation: direction, time window, trigger, confirmation, target/zone, stop or invalidation.	Undisclosed proprietary formulas or claims of certainty.
Progressive disclosure: show the simple view first, then evidence and assumptions.	Historical examples without complete denominator and point-in-time replay.
Clear reevaluation logic when a trigger fails or thesis changes.	AI-generated action that bypasses deterministic policy/review.

6. Indonesian-market calculations, valuation and simulation

6.1 Market mechanics as data

As of the current IDX trading-mechanism page, Regular and Cash Market trading uses 1 lot = 100 shares and five common price-fraction bands. Production MUST load the effective exchange rule version rather than copy these values into business logic. Negotiated-market behavior is modeled separately.

Previous price band	Tick / price fraction	Order-price normalization
< Rp200	Rp1	Buy limit rounds down to tick; sell limit rounds up to tick.
Rp200 to < Rp500	Rp2	Same effective-date rule lookup.
Rp500 to < Rp2,000	Rp5	Same effective-date rule lookup.
Rp2,000 to < Rp5,000	Rp10	Same effective-date rule lookup.
≥ Rp5,000	Rp25	Same effective-date rule lookup.

```
q_exec = lot_size * floor(q_desired / lot_size)
buy_limit = tick * floor(raw_buy_price / tick)
sell_limit = tick * ceil(raw_sell_price / tick)

# Never read lot_size or tick from constants.py.
# Resolve: market_rule(venue, board, instrument_type, effective_at).
```

6.2 Financial normalization

- Preserve original reported value, unit, currency, period and source mapping alongside normalized value.
- Normalize reported units with $\text{Value_IDR} = \text{Value_reported} \times \text{unit_multiplier} \times \text{FX}(\text{if needed})$.
- Derive quarterly flow metrics only from cumulative flow disclosures when appropriate (for example $Q2 = H1 - Q1$); never difference balance-sheet stock variables.
- Map restatements and corrections bitemporally; a later correction does not alter what an earlier decision knew.
- Separate reported earnings from normalized earnings and keep evidence-linked adjustments for non-recurring items.

6.3 Sector-specific valuation



Figure 8. Sector valuation router. Methods are explicit policy, not a model's free-form choice.

Sector	Primary methods	Required drivers	Known invalid shortcut
Banks	Residual income; justified P/B	ROE, cost of equity, NIM, credit cost, NPL/LaR, coverage, CASA/LDR, capital	EV/EBITDA as primary
Commodities	Asset NAV/DCF; mid-cycle multiples	Volume, benchmark price, FX, royalty, capex, reserve life	Perpetual peak commodity price

Sector	Primary methods	Required drivers	Known invalid shortcut
Consumer	DCF; P/E; EV/EBITDA	Volume, mix, pricing, CPI/input cost, distribution, working capital	Treating inflation as real growth
Telecom	SOTP; DCF; lease-consistent EV/EBITDA	ARPU, traffic, churn, spectrum, leases, capex, segment mix	Mixed lease-accounting basis
Infrastructure	Project DCF; SOTP	Tariff, volume, maintenance capex, concession, refinancing, delay	Terminal value beyond concession with no renewal probability
Property	RNAV; P/NAV; normalized earnings	Project value, remaining cost, collections, rent, debt, title	Full value for inaccessible land bank

6.4 Scenario and fair-value distribution

Each valuation produces not one “magic number” but a range under versioned assumptions. The scenario engine combines approved parameter shocks into Bull/Base/Bear outcomes. Probabilities must be separately calibrated; they are not automatically derived from valuation spread.

Scenario	Typical assumption pattern	Output
Bear	Lower volume/margins, higher discount rate, weaker FX, delayed catalyst	Fair value, return, downside, liquidity stress
Base	Management/consensus-like operating path with approved normalized assumptions	Primary fair-value range and expected net return
Bull	Upside operating case with explicitly bounded assumptions	Upside range; cannot erase base/bear risks

6.5 Corporate actions

Corporate-action logic is part of the return engine, not a display adjustment. Dividend, split, rights, bonus, merger/spin-off and symbol-change events must be represented explicitly and applied using then-effective dates.

```

TERP = (P_cum * N_old + P_subscription * N_new) / (N_old + N_new)

# Rights simulation must represent investor election:
# SUBSCRIBE | SELL_RIGHT | LAPSE
# Avoid double-counting dividend cash when using adjusted prices.

```

6.6 Point-in-time backtest and simulation



Figure 9. Point-in-time backtest timeline. The first executable session occurs after the decision snapshot; same-bar leakage is prohibited.

- Eligibility requires $\text{publication_time} + \text{ingestion_lag} \leq \text{decision_time}$.
- Universe includes then-listed, suspended, delisted, failed and special-board securities when applicable; never reconstruct history from today's survivors.
- Walk-forward parameters are frozen before the next evaluation window.
- Net performance includes effective broker/exchange/clearing/tax/stamp/spread/impact/tick costs as versioned parameters.
- Block bootstrap may be used for portfolio simulations to preserve serial and cross-sectional dependence; output must distinguish simulation from historical backtest.
- Minimum scorecard: gross/net CAGR, volatility, Sharpe/Sortino, max drawdown, hit rate, turnover, capacity, concentration, beta, tracking error, information ratio, tail loss, calibration and implementation shortfall.

7. Data sources, point-in-time pipeline and data contracts

7.1 Source tiers and licensing gates

Tier	Examples	Permitted role	Launch gate
M0	Yahoo Finance via yfinance (.JK)	FREE-ONLY EOD/delayed market-price and best-effort fundamentals for the current MVP.	No API key or paid fallback; personal/research use only until upstream terms and distribution rights are approved.
0	OJK, IDX rules/disclosures, KSEI, BI	Authority for rules, filings, events and macro facts.	Terms and automation rights reviewed.
1	Future licensed IDX market/reference feed	Not enabled in the current implementation; possible later production-grade source.	Separate commercial/legal approval before adapter activation.
2	Future commercial vendors	Not enabled while FREE_ONLY_MARKET_DATA=true.	Explicit ADR + budget + rights validation required before use.
3	Issuer IR and reputable research	Context and analyst evidence.	Reconciled to official timestamp and clearly labeled.

Free-only MVP rule

The runnable repository MUST set FREE_ONLY_MARKET_DATA=true and MARKET_DATA_PROVIDER=yfinance. The application contains no automatic paid-provider fallback. If the free upstream is unavailable, the run fails visibly instead of silently incurring cost.



Figure 10. Data pipeline. Raw source bytes are immutable; quality repair creates a new version rather than changing history.

7.2 Point-in-time and bitemporal semantics

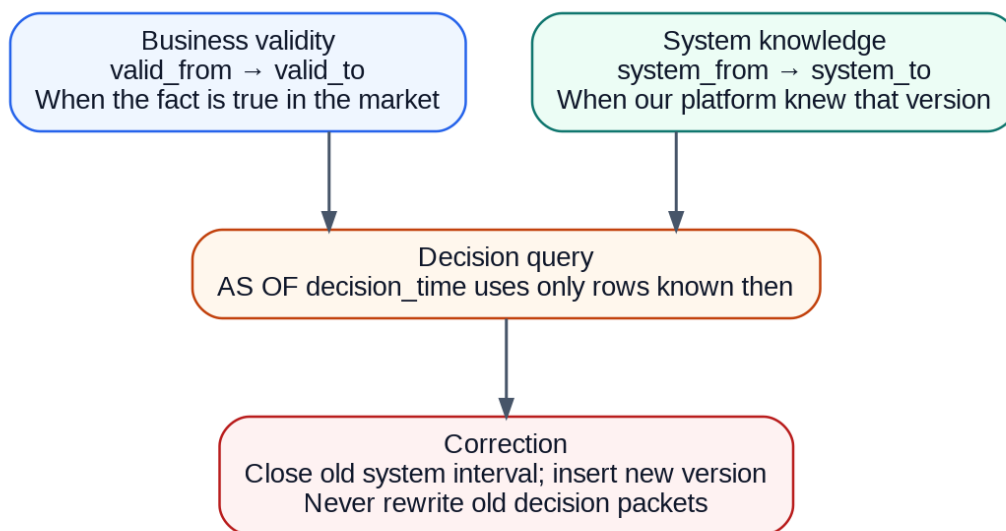


Figure 11. Two time axes are required: when a fact is valid and when the platform knew that fact.

7.3 Canonical observation contract

```
{
  "schema_version": "1.0",
  "instrument_id": "uuid",
  "venue": "XIDX",
  "observation_type": "DAILY_BAR",
```

```
"observed_at": "2026-09-18T09:00:00Z",
"effective_at": "2026-09-18T09:00:00Z",
"received_at": "2026-09-18T09:00:03Z",
"currency": "IDR",
"values": {"close": "1250", "volume": "1200000"},
"source_snapshot_id": "uuid",
"rule_version": "idx-equity-rule-2024-12",
"license_tags": ["NON_DISPLAY"],
"content_hash": "sha256:..."
}
```

7.4 Data quality controls

Control	Minimum behavior	Action on failure
Duplicate detection	Natural/business key + source event ID + checksum.	Quarantine duplicate set; do not silently drop if values conflict.
Referential integrity	100% instrument/source/version links.	Block snapshot sealing.
Timestamp normalization	UTC storage + explicit Asia/Jakarta market context.	Reject ambiguous local timestamps.
Corporate-action reconciliation	Material events reconcile raw → adjusted price/share/cash path.	Quarantine affected instrument.
Freshness	Dataset-specific max age by audience/action.	Mark stale; block actionable publication if hard limit breached.
Schema evolution	Backward/forward compatibility rules for event/data contracts.	Route incompatible payload to dead-letter quarantine.
License tags	Every source snapshot carries rights tags and retention/export obligations.	Deny query/export when policy cannot prove entitlement.

8. Target architecture and component design

Architecture choice

Preferred MVP runtime: contract-enforced modular monolith for workflow-heavy domains, plus isolated ingestion and compute workers. Keep public schemas, IDs, event contracts, evidence format and audit semantics stable so services can be extracted later without rewriting the product model.

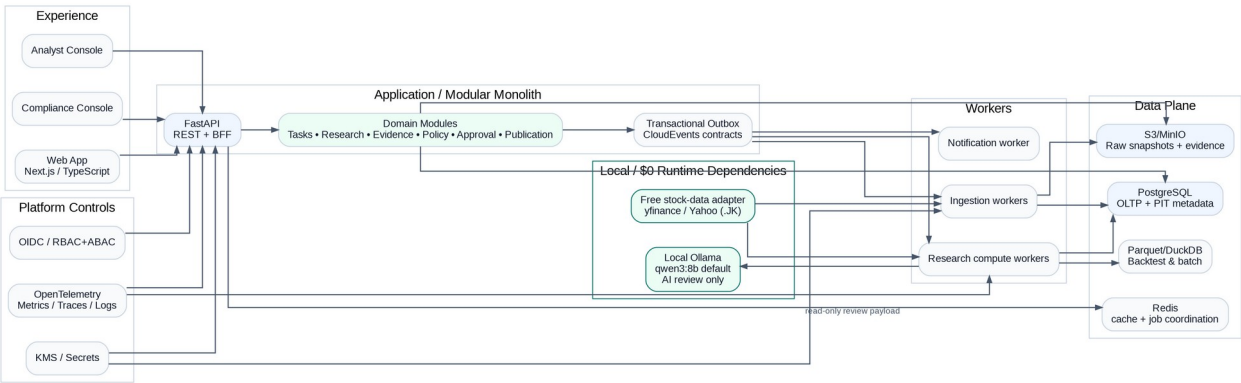


Figure 12. Target architecture. The application is simple enough to ship, while data and compute workloads remain isolated.

8.1 Recommended implementation stack

Layer	MVP choice	Why	Exit / scale path
Web	Next.js + TypeScript	Fast product iteration, SSR where useful, typed contracts.	Separate mobile client can reuse OpenAPI.
API	Python 3.12 + FastAPI + Pydantic	Natural fit for financial/quant code and contract-first REST.	Extract high-load domains behind same schemas.

Layer	MVP choice	Why	Exit / scale path
Market data	Free-only yfinance adapter (Yahoo public/unofficial endpoints)	No API key/per-call charge; broad .JK coverage for EOD prototype.	Replace only through MarketDataProvider interface after rights/budget approval.
AI review	Local Ollama + qwen3:8b default	No cloud AI key or per-token bill; data stays local; AI remains non-authoritative.	Model is configurable through OLLAMA_MODEL; cloud provider is not implemented in MVP.
ORM/migrations	SQLAlchemy 2 + Alembic	Explicit transactions, mature PostgreSQL support.	Keep migrations owned by bounded module.
Workflow/jobs	PostgreSQL outbox + Redis-backed worker queue	Low operational overhead for MVP; durable event intent in DB.	Kafka-compatible log or Temporal when throughput/workflow complexity justifies.
OLTP/PIT metadata	PostgreSQL	Strong consistency, constraints, row-level security.	Partition, read replicas, domain extraction.
Time-series/batch	PostgreSQL/Timescale-compatible tables + Parquet/DuckDB for research	Good MVP economics and reproducible offline analytics.	ClickHouse/lakehouse if data scale requires.
Object evidence	S3-compatible object store	Immutable raw/evidence blobs with versioning.	Cross-region replication + object lock.
Cache	Redis	Disposable cache + job coordination.	Cluster only when needed.
Auth	OIDC provider + RBAC/ABAC in app	Enterprise-friendly identity; policy remains in domain.	Dedicated authorization service if tenancy grows.
Observability	OpenTelemetry + metrics/logs/traces backend	Portable instrumentation and correlation IDs.	Managed backend interchangeable.

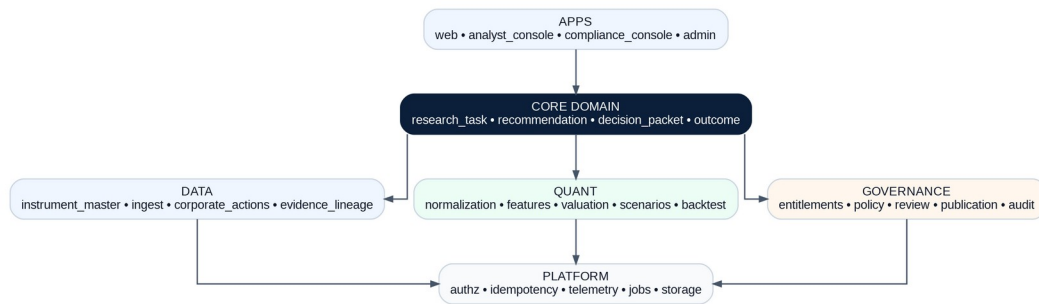


Figure 13. Source-code module map. Dependency direction points toward the core domain, not from domain to infrastructure.

8.2 Repository blueprint

```

buktiSaham/
├── apps/
│   ├── web/                # Next.js user-facing research workspace
│   ├── analyst_console/    # analyst authoring/review UI
│   └── compliance_console/  # policy, approval, audit UI
├── services/
│   ├── api/                # FastAPI composition root
│   ├── worker_ingest/      # source adapters + data quality
│   └── worker_research/    # features, valuation, scenarios, backtests
├── packages/
│   ├── domain/             # entities, value objects, state machines
│   ├── research_engine/    # pure deterministic quant functions
│   ├── evidence/           # lineage + claim binding
│   ├── policy/             # versioned decision rules
│   ├── contracts/          # OpenAPI, JSON Schema, event schemas
│   └── platform/           # authz, idempotency, telemetry, storage
├── data/
│   ├── adapters/           # free yfinance + future official/vendor adapters
│   ├── pipelines/          # PIT normalization and snapshot sealing
│   ├── quality/            # reconciliation + quarantine rules
│   ├── fixtures/           # golden PIT datasets for tests
│   └── policies/           # YAML/JSON versioned policy parameters
├── infra/                  # Docker, IaC, environment modules
├── ops/runbooks/           # incident + recovery procedures
├── security/threat_models/
├── docs/adr/               # architecture decision records
└── tests/{unit,property,contract,integration,e2e,quant,security,resilience}/
  
```

8.3 Component responsibilities

Component	Owns	Must not own
Research Task	Intent, parameters, as-of, run creation.	Valuation math or publication approval.
Instrument Master	Effective symbols, boards, sectors, listing status, identifiers.	Recommendation state.
Market Ingest	Source retrieval, raw snapshots, parsing, receipt time.	Business interpretation of valuation.
Corporate Actions	Event-sourced actions and reconciliation factors.	UI presentation logic.
Research Engine	Pure financial normalization, features, valuation, scenario, execution metrics.	Authorization or mutable persistence.
Evidence Lineage	Claim → evidence mapping, transforms, hashes, export index.	Free-form claim generation.
Policy	Eligibility, freshness, audience, thresholds, action rules.	Source ingestion.
Review/Approval	Human roles, comments, approval state transitions.	Editing sealed quantitative output.
Publication	Seal packet, enforce expected version, distribute permitted content.	Creating research outputs.
Outcome Evaluator	Score complete issued population after horizon/invalidation.	Retrospectively changing action labels.

9. Domain model, database schema and data retention

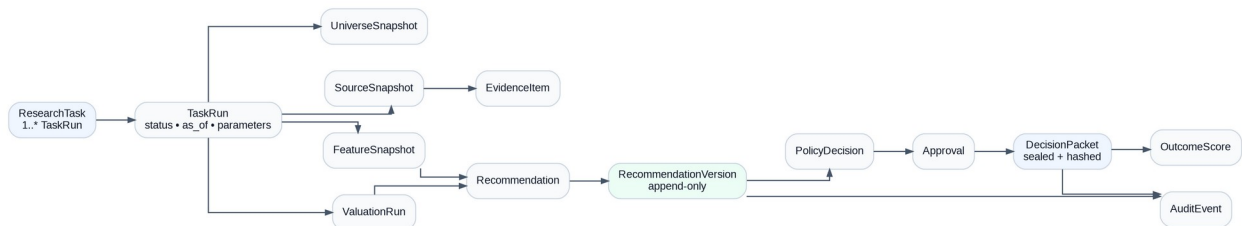


Figure 14. Core domain model. RecommendationVersion and DecisionPacket are append-only audit anchors.

9.1 Core entity invariants

Entity	Key invariant
ResearchTask	Parameters are versioned; task identity survives reruns.
TaskRun	One immutable parameter snapshot per run; stage events are append-only.
SourceSnapshot	Raw content checksum and rights tags never change.
EvidenceItem	Evidence is addressable by stable ID and binds to source snapshot + transform.
FeatureSnapshot	All feature values declare formula/version and as-of context.
ValuationRun	Method, inputs, assumptions, sensitivity grid and model version are stored together.
RecommendationVersion	Append-only; supersedes_id links revisions; sealed numeric payload immutable.
PolicyDecision	Records policy version, inputs, reason codes and pass/fail details.
Approval	Actor, role, decision, timestamp and scope; cannot be overwritten.
DecisionPacket	Hash over exact published payload + evidence index + approvals.
OutcomeScore	References the exact issued version; no cherry-picked omission.
AuditEvent	Append-only security/governance trail with correlation/causation IDs.

9.2 Minimum PostgreSQL table set

Table	Primary purpose	Key fields / indexes
instrument	Stable instrument identity.	id, venue, isin, instrument_type; unique identifiers.
instrument_version	Effective-dated symbol/board/sector/status.	instrument_id, valid_from/to, system_from/to; GiST/range index.
market_rule_version	Lots, ticks, sessions, auto-rejection metadata.	rule_set, effective range, payload_json, source_snapshot_id.
source_snapshot	Immutable source retrieval record.	id, source_type, uri, observed/effective/received, checksum, license_tags.
financial_fact	Reported + normalized PIT financial facts.	instrument_id, concept, period, values, source/effective/system times.
corporate_action	Event-sourced actions.	event_type, key dates, terms, status, source_snapshot_id.
research_task	User intent.	id, tenant_id, task_version, params_json, as_of.
task_run	Asynchronous run state.	id, task_id/version, status, snapshot_hash, idempotency_key.
feature_snapshot	Versioned derived features.	run_id, feature_name/version, numeric_value, lineage_json.
valuation_run	Sector model results.	run_id, method, version, fair_value, sensitivity_json.
recommendation_version	Research action and scenario payload.	recommendation_id, version, supersedes_id, status, sealed_payload_hash.
policy_decision	Eligibility/action policy evaluation.	recommendation_version_id, policy_version, result, reasons_json.
approval	Review decisions.	recommendation_version_id, actor, role, decision, created_at.
decision_packet	Published immutable packet.	recommendation_version_id, content_hash, evidence_manifest_uri, published_at.
outcome_score	Performance/calibration record.	packet_id, horizon, return_gross/net, drawdown, status.
audit_event	Append-only audit.	event_id, actor/workload, action, aggregate_id, correlation_id, payload_hash.

9.3 Retention classes

Class	Examples	Default posture
R1 Immutable evidence	Raw official/licensed snapshots, decision packets, audit exports	Retain per contract/regulatory policy; object versioning; deletion only by governed legal workflow.
R2 Operational records	Tasks, runs, approvals, policy decisions	Retain to support audit/outcome history; archive after active lifecycle.
R3 Rebuildable analytics	Caches, search index, some derived aggregates	May be rebuilt from authoritative data; shorter retention allowed.
R4 Restricted licensed data	Dataset-specific market/reference records	Retention/export/cloud/AI use strictly per entitlement matrix.
R5 PII/account data	User identity and access metadata	Minimize; separate from market research data; privacy retention schedule.

10. API, event and workflow contracts

10.1 API principles

- OpenAPI is normative. Client SDKs are generated from the contract, not hand-maintained.
- All commands require actor context, tenant, correlation ID and Idempotency-Key.
- State-changing commands carry expected aggregate version for optimistic concurrency.
- Queries return freshness, provenance summary and entitlement-filtered content.

- Monetary values are decimal strings; timestamps are RFC3339 UTC with market timezone metadata where relevant.
- All enums include UNKNOWN for forward compatibility.

Method	Path	Purpose	Key errors
POST	/v1/research-tasks	Create task and initial version.	400 validation; 403 entitlement; 409 idempotency conflict
POST	/v1/research-tasks/{id}/runs	Start analysis from a specific task version.	409 version; 422 stale/unsupported input
GET	/v1/runs/{id}	Stage status, errors, hashes and result pointer.	403; 404
GET	/v1/instruments	Effective-dated eligible instrument search.	403 entitlement; 409 stale master search
GET	/v1/recommendations/{id}/versions/{v}	Read exact immutable version.	403 entitlement; 404
POST	/v1/recommendations/{id}/submit	Run policy and start review.	403 role; 409 version; 412 policy dependency
POST	/v1/recommendations/{id}/approve	Record independent approval.	403 role; 409 state/version
POST	/v1/recommendations/{id}/publish	Seal and distribute approved version.	409 approval/version; 451 restricted
POST	/v1/recommendations/{id}/invalidate	Create governed invalidation.	403 role; 409 version/state
GET	/v1/audit/exports/{id}	Retrieve signed evidence package.	403 auditor; 425 assembling

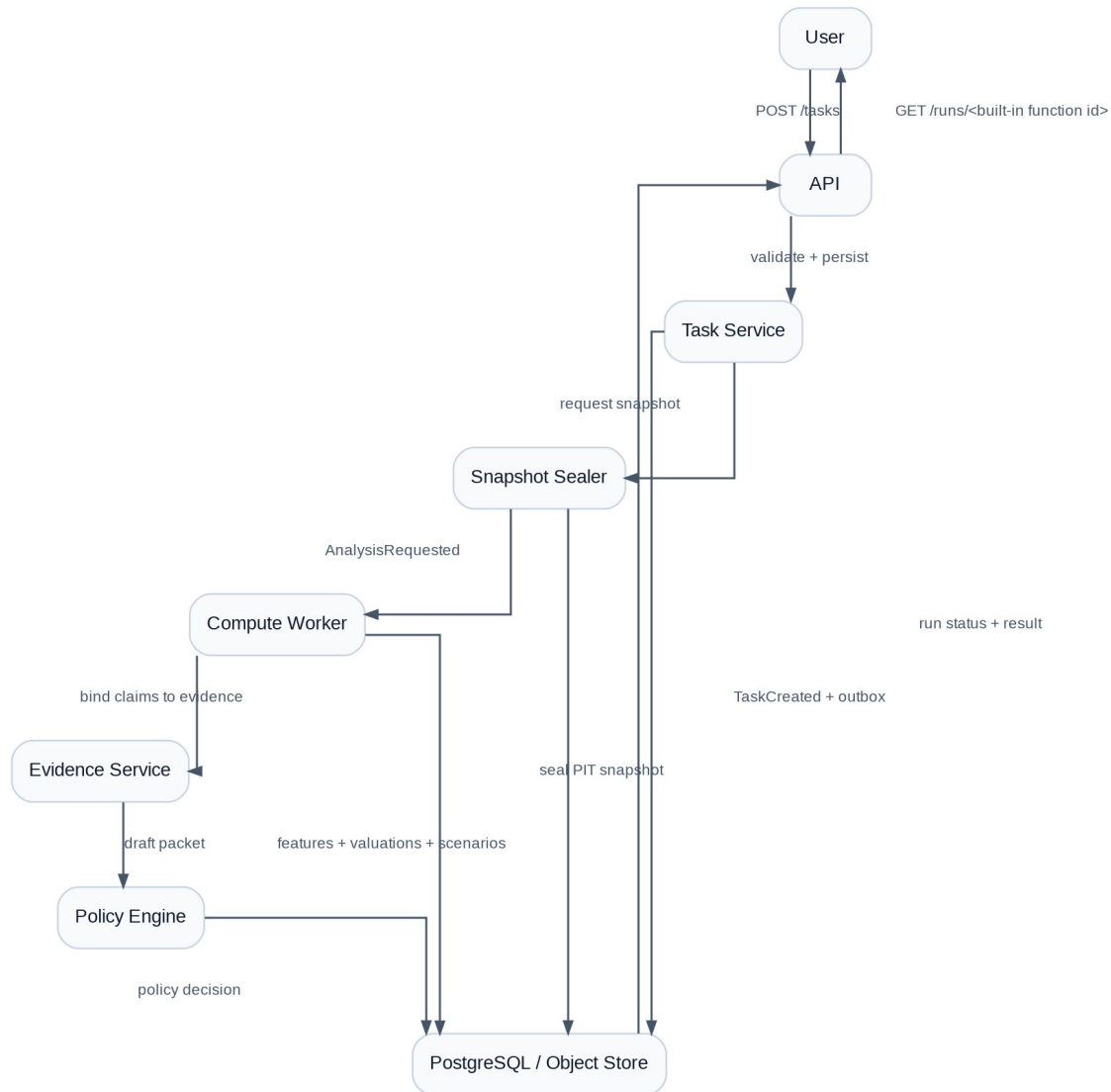


Figure 15. Run-analysis sequence. Snapshot sealing happens before compute.



Figure 16. Publish sequence. Human review is downstream of deterministic policy and upstream of immutable sealing.

10.2 Event contracts

Events use a CloudEvents-aligned envelope. At MVP they can be persisted in a transactional outbox and dispatched to workers; the schema remains compatible with a future durable event log.

```

{
  "event_id": "uuid",
  "type": "RecommendationPublished",
  "schema_version": "1.0",
  "occurred_at": "...Z",
  "producer": "publication",
  "tenant_id": "uuid",
  "aggregate": {"type": "Recommendation", "id": "uuid", "version": 7},
  "correlation_id": "uuid",
  "causation_id": "uuid",
  "classification": "INTERNAL",
  "payload": {"decision_packet_id": "uuid"},
  "payload_hash": "sha256:..."
}

```

Core event names: SourceSnapshotSealed, InstrumentMasterUpdated, CorporateActionReconciled, AnalysisRequested, FeatureSetComputed, ValuationCompleted, RecommendationDrafted, PolicyCheckCompleted, RecommendationSubmitted, ApprovalRecorded, RecommendationPublished, RecommendationRevised, RecommendationInvalidated, RecommendationExpired and OutcomeScored.

11. AI usage and guardrails

Authority boundary

The MVP AI runtime is local Ollama (qwen3:8b by default). AI is an optional explanation layer. It may summarize evidence, generate plain-language comparisons, surface contradictions and draft analyst notes. It **MUST NOT** change market data, financial facts, valuation numbers, scenario probabilities, policy outcomes, approval state or publication action.

Use case	Allowed?	Control
Summarize a filing already in evidence store	Yes	Only approved evidence IDs; citations mandatory; output schema validated.
Explain why fair value changed between versions	Yes	Input is machine-generated diff; no new numbers.
Suggest questions for analyst review	Yes	Clearly labeled suggestion; cannot block/publish.
Generate a recommendation action from free-form prompt	No	Recommendation action is deterministic policy output.
Retrieve arbitrary internet source during publish	No	Publish uses approved evidence sources only.
Use any external/cloud model API in the MVP	No	The implemented MVP uses local Ollama only; no external AI API adapter or key is required.

11.1 Local Ollama runtime contract

```
AI_REVIEW_ENABLED=true
OLLAMA_BASE_URL=http://ollama:11434
OLLAMA_MODEL=qwen3:8b

FREE_ONLY_MARKET_DATA=true
MARKET_DATA_PROVIDER=yfinance
```

- The worker sends only deterministic analysis fields and approved evidence summaries to the local Ollama REST API.
- Ollama output is stored as review metadata and cannot overwrite research_action, numeric metrics, evidence hashes or policy results.
- If Ollama is unavailable, the deterministic run still completes with ai_review.status = unavailable.
- Model download is explicit through scripts/pull-model.sh; changing OLLAMA_MODEL is a reversible runtime configuration change.

11.2 Prompt/tool security

- Model registry with approved provider/model/version and kill switch.
- Evidence and tool allowlists; no arbitrary URL/tool access inside publish workflow.
- Prompt injection test corpus across filings, web content and analyst notes.
- Structured response schema validation before text is displayed.
- No secrets, PII or restricted raw data in prompts unless explicitly permitted by classification policy.
- Store prompt template version, evidence IDs, model ID and output hash for reproducibility of narrative context.

12. Security, compliance and entitlement controls

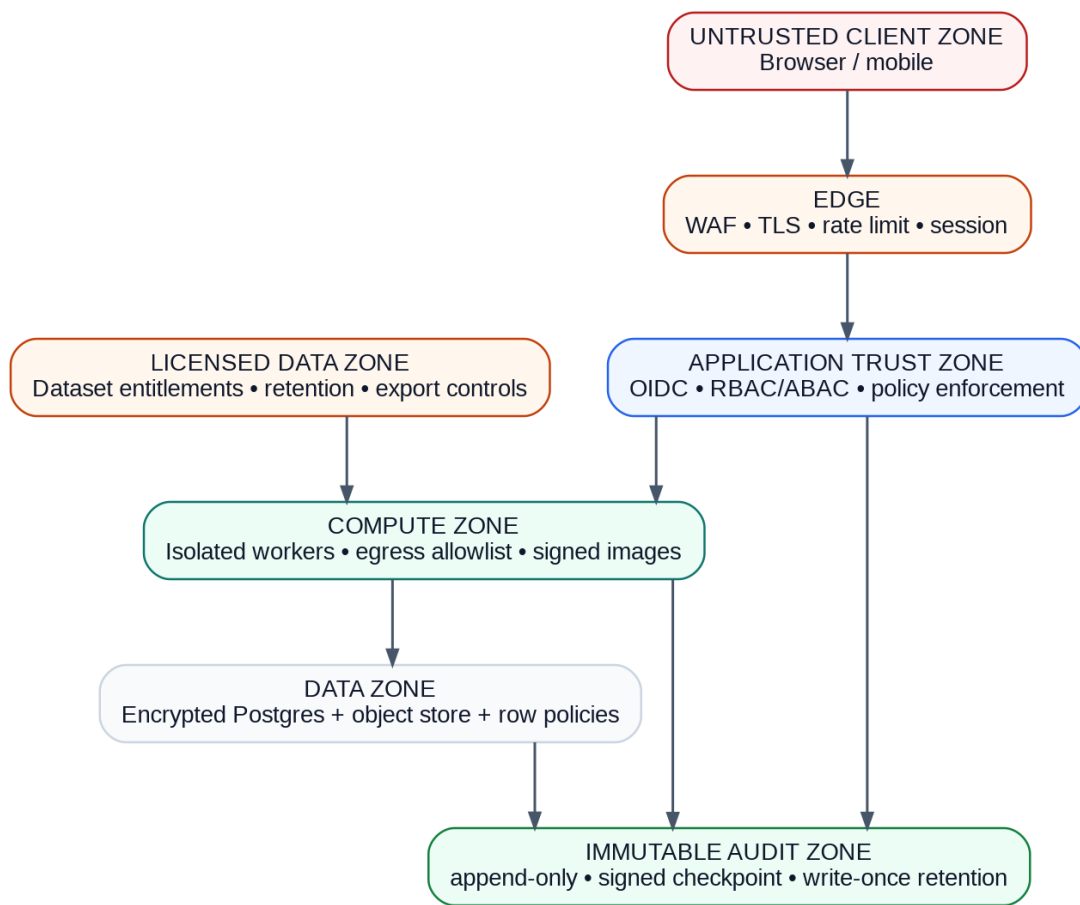


Figure 17. Security trust boundaries. Licensed data and audit evidence are separate control planes.

12.1 Authorization model

RBAC is necessary but not sufficient. Authorization also evaluates tenant, audience, instrument, dataset license, geography, data classification, action, export intent and where relevant suitability status. Policy should be centralized as testable domain rules, even if identity comes from an external OIDC provider.

Role	Typical permissions	Explicit denial
Reader	Read published/authorized packets.	No raw restricted export, no workflow mutation.
Analyst	Create tasks, run analysis, draft/revise recommendations.	Cannot independently approve own draft.
Reviewer	Review and request changes.	Cannot modify deterministic results.
Publisher/Compliance	Approve/publish within scoped policy.	Cannot bypass failed hard policy gate.
Data Steward	Quarantine/repair data, seal corrected versions.	Cannot retroactively mutate published packet.
Auditor	Read audit exports and history.	No authoring.
Operator	Operate platform, feature flags, incident response.	No content approval unless separately assigned role.

12.2 Compliance/data-rights gating

- Legal classifies research vs advice vs alerting vs personalization for the actual operating entity and target audience before public launch.
- Data governance maintains an entitlement matrix for display, non-display, storage, archive, cloud processing, derived analytics, AI/model use, API export, redistribution and deletion for every dataset.
- Publication policy verifies audience and dataset rights at request time, not only at ingest.

- All user-facing claims distinguish sourced fact, derived calculation and editorial judgment.
- Broker handoff remains disabled until suitability, execution, surveillance, recordkeeping and partner responsibilities are separately approved.

12.3 Security controls

Control family	Requirement
Identity	Phishing-resistant MFA for privileged roles; short sessions for admin; separate break-glass identities.
Secrets	KMS/HSM-backed encryption; automatic rotation; no secret in code, prompt, event or log.
Network	Private data services; controlled egress; WAF/rate limiting at edge; service-to-service identity.
Data	Encryption in transit/at rest; row/tenant policy; classification tags; object versioning and retention.
Build	Branch protection, signed commits/build provenance where feasible, SAST/SCA/secret scan/SBOM/container/IaC scan.
Audit	Append-only event ledger, signed checkpoints, immutable evidence export; operational logs are supplementary, not the only audit record.
AI	Model allowlist, prompt/tool/data policies, injection testing, schema validation, human publish approval, kill switch.

13. Non-functional requirements and SLOs

Indicator	Initial target	Measurement	Breach action
Published recommendation read availability	99.95% monthly	Successful authorized reads / valid requests	Hold rollout; page owner; serve labeled cache if safe.
Authoring/admin availability	99.9% monthly	Successful workflow ops	Degrade to read-only.
Recommendation read latency	p95 ≤ 300 ms	API edge; cache status segmented	Serve cache; shed non-critical enrichment.
Evidence page latency	p95 ≤ 2 s	Full lineage retrieval	Disable heavy previews; keep source index.
Licensed-feed ingest lag	Dataset-specific; target p99 ≤ 5 s only for licensed real-time feed	Source timestamp → canonical receipt	Mark stale; block actionability when threshold crossed.
Evidence completeness	100% mandatory publish fields	Packet validator	Fail publication.
Run orchestration	99% accepted runs reach terminal state	Run state machine	Alert stuck runs; safe retry.
Audit write durability	No acknowledged publish without durable audit/packet record	Transactional checks	Fail closed.

13.1 Capacity assumptions

- MVP design target: selected liquid IDX equities, EOD/delayed research, tens to low hundreds of concurrent users, background analysis rather than HFT.
- Before controlled beta, load test at 10× forecast read peak and 3× forecast ingest/compute peak.
- Partition large observation/feature tables by effective date and/or instrument hash as measured scale requires.
- Cache is an optimization only; all authoritative results must be reconstructable without cache.
- Backtests run in separate worker pool and storage path so research exploration cannot starve publication or user read paths.

13.2 Accessibility and internationalization

- WCAG 2.2 AA target: keyboard operation, screen-reader chart summaries, focus states and non-color-only risk cues.
- Primary UX Bahasa Indonesia with English technical terms only where clearer; contracts and code remain English.
- Currency/number/date formatting is locale-aware but storage is canonical decimal + ISO/RFC timestamps.
- All risk and evidence labels must remain meaningful when exported to PDF/CSV or viewed without color.

14. Testing and quality strategy

Layer	Scope	Release blockers
Unit	Money/decimal, date/calendar, lot/tick, financial normalization, valuation math.	Any deterministic failure.
Property-based	Probability sums, monotonic valuation relationships, state-machine invariants, round-trip adjustments.	Invariant counterexample.
Golden data	Known issuer filings, corporate actions, market rules and PIT snapshots.	Reconciliation outside tolerance.
Contract	OpenAPI, JSON Schema, event compatibility, idempotency, out-of-order delivery.	Breaking unversioned contract.
Integration	DB transactions, outbox/inbox, object store, auth, worker orchestration.	Data loss/duplication or unauthorized access.
Quant	Leakage, survivorship, walk-forward, costs, drawdown, capacity, calibration/Brier score, abstention quality.	Material methodology breach.
Security	Tenant isolation, authz matrix, prompt injection, exfiltration, secret handling, break-glass.	Any critical/high unresolved issue.
Resilience	Feed gaps, corrupt event, queue/db/object-store failure, clock skew, replay/restore.	Cannot fail closed or recover within target.
UX/compliance	Source/freshness/risk/invalidation comprehension and prohibited claim corpus.	<90% comprehension or any policy escape.

14.1 Definition of Done for each P0 story

- 1. Behavior implemented behind a contract and feature flag where applicable.
- 2. Unit/property tests cover happy path and boundary conditions.
- 3. Golden PIT fixture proves deterministic output.
- 4. Authorization and entitlement tests cover allow and deny paths.
- 5. Observability includes correlation ID, domain metrics and meaningful structured logs.
- 6. Failure mode is defined and user-facing message is non-misleading.
- 7. Migration/rollback path documented.
- 8. Product, engineering and relevant data/quant/compliance owner accept the story's criteria.

15. Deployment, observability and disaster recovery

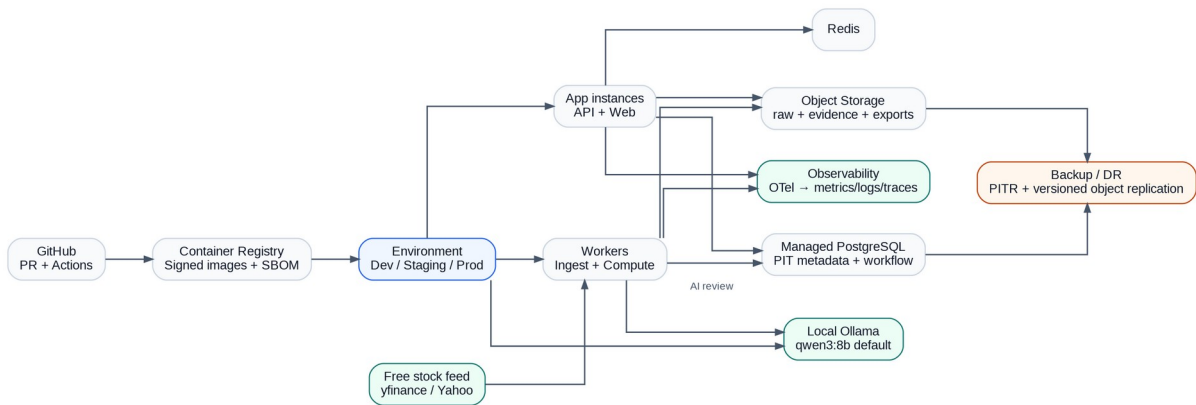


Figure 18. Deployment topology. Production data and evidence are durable; cache/search remain rebuildable.



Figure 19. CI/CD pipeline. Security, contract and PIT replay tests run before a human production approval.

15.1 Environment strategy

Environment	Data	Purpose
Local	Synthetic + curated fixtures only	Developer productivity; Docker Compose; deterministic offline tests.
CI	Ephemeral fixture datasets	Automated tests, schema compatibility, migrations.
Staging	Production-like but non-production entitlements/data rights	End-to-end, security, load, replay and UAT.
Production/private MVP	Free yfinance adapter only while FREE_ONLY_MARKET_DATA=true; official evidence where automation rights permit	Private/research deployment only; public redistribution remains gated by upstream/IDX rights review.

15.2 Observability and incident response

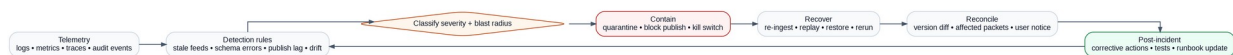


Figure 20. Incident loop. Data/evidence issues can automatically stop publication before they become user harm.

- Every request/run/event carries correlation_id; recommendation/version IDs appear in traces and structured logs.
- Alerts: feed gaps, stale instruments, schema failures, evidence completeness, policy dependency outage, approval latency, publish lag, outcome calibration drift and anomalous export.
- Recommended recovery objectives: recommendation metadata RPO ≤5 min / RTO ≤60 min; immutable evidence/audit near-zero RPO via object versioning/replication / RTO ≤4 h; rebuildable search/analytics RPO/RTO ≤24 h.
- Quarterly restore tests, semiannual failover rehearsal and annual cyber-recovery exercise record actual evidence, not only written procedure.

16. Delivery roadmap, backlog and ownership



Figure 21. Delivery roadmap. Audience expansion only follows evidence, legal/data rights and operational gates.

16.1 Phases and exit gates

Phase	Core deliverable	Exit gate	Rollback posture
0 Governance foundation	Legal perimeter, data RFP/rights matrix, taxonomy, threat model, source contracts.	Named approvers sign data, retention, publication and audience boundaries.	No production data flow.
1 Research kernel	PIT ingest, instrument master, corporate-action engine, deterministic normalization/valuation/backtest.	Replay + reconciliation pass on representative historical samples.	Delete non-required vendor samples per contract.
2 Analyst MVP	Task workflow, recommendation card, evidence drawer, policy/review, immutable packet.	100% evidence completeness; shadow calibration approved.	Feature flags disable generation/publish.
3 Controlled paper beta	Limited users/universe, delayed data if licensed, scorecards and alerts.	SLO, comprehension, security and incident drills pass.	Traffic to 0%; retain audit.
4 Invited distribution	Broader web/mobile distribution, daily compliance review, monitoring.	Independent control assurance and penetration test.	Audience/instrument kill switches.
5 GA	General availability; optional separately approved integrations.	Public methodology, complete outcome ledger, DR exercise and all legal/data gates.	Read-only archive + withdrawal notices.

16.2 Implementation epics

Epic	Name	Implementation result
E1	Foundation contracts	OpenAPI baseline, domain IDs, event envelope, money/time library, ADRs

Epic	Name	Implementation result
E2	Identity + entitlement	OIDC, tenant model, RBAC/ABAC, dataset rights matrix, audit
E3	Instrument master	Effective-dated symbols/status/sector/board + source reconciliation
E4	Raw ingest	Adapters, immutable object store, source snapshot registry, checksum
E5	PIT canonical model	Bitemporal observations/financial facts/rules, schema versioning
E6	Corporate actions	Event model, KSEI/issuer reconciliation, adjustment engine
E7	Research engine	Normalization, features, technical context, local execution math
E8	Valuation	Sector router + model implementations + sensitivities
E9	Scenario engine	Versioned assumptions, probabilities, expected net return, risk metrics
E10	Evidence lineage	Claim registry, transform lineage, content hash, evidence drawer APIs
E11	Policy engine	Freshness, rights, threshold/action rules, fail-closed logic
E12	Review + approval	State machine, comments, independent approval, version diff
E13	Publication	Immutable packet seal, disclaimer/version, distribution, withdrawal
E14	Outcome evaluator	Horizon/invalidation scoring, population reconciliation, calibration
E15	Web experience	Task wizard, run progress, card, evidence drawer, comparisons
E16	Operations	OTel, dashboards, alerts, runbooks, backup/restore, DR
E17	Security hardening	Threat model, SAST/SCA, secrets, WAF, penetration test
E18	AI explanation layer	Evidence-bound summarization, schema validation, injection tests, kill switch

16.3 Ownership and irreversible decisions

Workstream	Accountable role	First irreversible decision	Required sign-off
Product/research policy	Product Lead	Public taxonomy and research-view semantics	Product, Compliance, Legal
Data licensing + PIT	Data Lead	Licensed storage/derived-use model and time semantics	Data, Legal, Security
Quant methodology	Quant Lead	Published scoring/valuation policy version	Model Risk, Compliance
Architecture/contracts	Engineering Lead	Canonical IDs, public schemas, packet/audit format	Architecture, Security
Security/operations	Security Lead	Key hierarchy and audit trust root	Security, Risk, Operations

17. Approval checklist and handoff to implementation

Implementation may start when the following are explicitly approved:
Product scope and action taxonomy; legal research/advice boundary; initial allowed audience; data-source/rights plan; point-in-time model; architecture D-01–D-08; sector valuation coverage; recommendation policy framework; security model; SLO targets; phase-1/2 acceptance gates.

17.1 Approval checklist

Area	Approval condition
Product	Primary journey, P0 scope and non-goals accepted.
Legal/Compliance	Operator/audience/research-vs-advice boundary accepted; broker handoff remains off.
Data	Representative source samples and rights matrix available.
Quant	Sector methods, cost model, scenario framework and validation owner named.
Engineering	Architecture, repository boundaries, DB/API/event contracts accepted.
Security	Threat model, authz, secret/key, audit and incident requirements accepted.
Operations	SLOs, RPO/RTO, monitoring, restore/failover test plan accepted.
UX	Recommendation card/evidence drawer comprehension criteria accepted.

17.2 Implemented MVP handoff and next sprint

Current runnable baseline
The companion ZIP already implements Docker Compose, FastAPI, PostgreSQL, Redis/RQ, the free-only yfinance market adapter, deterministic scoring/scenarios/policy, a local Ollama review client, Next.js UI, tests and operational scripts. The next sprint should deepen point-in-time official evidence and sector valuation rather than replace these boundaries.

1. Create monorepo skeleton and CI pipeline with lint/type/unit/contract/security gates.
2. Implement money/time/decimal and effective-date primitives plus test fixtures.
3. Create PostgreSQL baseline schema: tenant, actor, instrument, instrument_version, source_snapshot, research_task, task_run, audit_event.
4. Publish OpenAPI v1 draft for task creation/run status and generate typed clients.
5. Implement transactional outbox + idempotency middleware + correlation IDs.
6. Build synthetic/fixture instrument-master and market-rule adapters before real licensed feeds.
7. Implement snapshot-sealing service and content-hash manifest.
8. Ship minimal web task wizard + run progress page using fixture data.
9. Add one end-to-end golden test: create task → seal snapshot → compute stub valuation → draft packet → policy preview.
10. Freeze ADRs for PIT semantics, immutable recommendation versioning and effective-dated market rules.

Appendix A. API examples

A.1 Create Research Task

```
POST /v1/research-tasks
Idempotency-Key: 8f7...

{
```

```

"title": "BBRI 6M valuation review",
"universe": {"type": "symbols", "values": ["BBRI"]},
"horizon": "P6M",
"as_of": "2026-09-18T09:00:00Z",
"capital": {"currency": "IDR", "amount": "100000000"},
"mandate": {"max_position_pct": "0.20", "max_bear_downside": "0.25"},
"cost_profile_id": "id-retail-standard-v1",
"entitlement_profile_id": "retail-delayed-v1",
"audience": "RESEARCH_USER"
}

```

A.2 Recommendation payload shape

```

{
  "recommendation_id": "uuid",
  "version": 3,
  "instrument": {"symbol": "BBRI", "venue": "XIDX"},
  "as_of": "2026-09-18T09:00:00Z",
  "horizon": "P6M",
  "research_view": "WATCH",
  "status": "IN_REVIEW",
  "scenarios": [
    {"name": "BEAR", "probability": "0.25", "fair_value": "..."},
    {"name": "BASE", "probability": "0.50", "fair_value": "..."},
    {"name": "BULL", "probability": "0.25", "fair_value": "..."}
  ],
  "metrics": {
    "expected_net_return": "...",
    "bear_downside": "...",
    "confidence": "...",
    "evidence_completeness": "..."
  },
  "valuation_methods": ["RESIDUAL_INCOME", "JUSTIFIED_PB"],
  "triggers": [...],
  "invalidations": [...],
  "evidence_ids": ["uuid", "uuid"],
  "policy": {"version": "policy-2026-09-shadow-01", "reasons": [...]},
  "snapshot_hash": "sha256:..."
}

```

Appendix B. Database table blueprint

B.1 Recommendation version (illustrative DDL)

```

CREATE TABLE recommendation_version (
  id UUID PRIMARY KEY,
  recommendation_id UUID NOT NULL,
  version INTEGER NOT NULL,
  supersedes_id UUID NULL REFERENCES recommendation_version(id),
  task_run_id UUID NOT NULL REFERENCES task_run(id),
  instrument_id UUID NOT NULL REFERENCES instrument(id),
  status TEXT NOT NULL,
  research_view TEXT NOT NULL,
  horizon TEXT NOT NULL,
  as_of TIMESTAMPTZ NOT NULL,
  scenario_payload JSONB NOT NULL,
  metric_payload JSONB NOT NULL,
  valuation_payload JSONB NOT NULL,
  snapshot_hash TEXT NOT NULL,
  sealed_payload_hash TEXT NULL,
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
  UNIQUE (recommendation_id, version)
);

-- Application policy: rows linked to a published DecisionPacket are immutable.
-- Revision inserts a new row and sets supersedes_id.

```

B.2 Market rule version (illustrative DDL)

```

CREATE TABLE market_rule_version (

```

```

id UUID PRIMARY KEY,
venue TEXT NOT NULL,
board TEXT NOT NULL,
instrument_type TEXT NOT NULL,
rule_kind TEXT NOT NULL, -- LOT_SIZE | TICK_TABLE | SESSION | COST | ...
valid_from TIMESTAMPTZ NOT NULL,
valid_to TIMESTAMPTZ NULL,
system_from TIMESTAMPTZ NOT NULL DEFAULT now(),
system_to TIMESTAMPTZ NULL,
payload JSONB NOT NULL,
source_snapshot_id UUID NOT NULL REFERENCES source_snapshot(id),
content_hash TEXT NOT NULL
);

```

Appendix C. Reference and verification register

The implementation design intentionally treats regulations, trading mechanics, taxes and licenses as effective-dated inputs. The following references are design anchors, not permission to reuse data. Legal counsel and licensors must verify the exact 2026 terms, operator entity and intended audience before launch.

Reference family	How it is used
Uploaded source blueprint	Implementation Blueprint for an Explainable Indonesian Equity Research Platform, 19 Sep 2026. Used as the reference baseline for evidence-first design, point-in-time lineage, controlled publication, modular architecture and launch gating.
IDX	Trading Hours and Mechanism; Rule II-A / current decree for equity trading. Verified current public page states 1 lot = 100 shares in Regular/Cash markets and five common price-fraction bands; production must use effective rule versions.
OJK	POJK 29/POJK.04/2016 on Annual Reports of Issuers/Public Companies; POJK 14/POJK.04/2022 on Periodic Financial Statements; material disclosure and consumer-protection regulations as applicable.
KSEI	Corporate Action process, announcements and schedules. Used as an event/reconciliation source candidate subject to automation rights.
Bank Indonesia	JISDOR/FX methodology, inflation target and relevant rate/macro publications. Used as point-in-time macro inputs, not hard-coded forecasts.
Astronacci	Eye of Future public material. Used only as a presentation reference for scenario/trigger/invalidation grammar; no proprietary formula is assumed.
OpenAPI Initiative	OpenAPI Specification. Normative REST contract format.
CNCF	CloudEvents Specification. Event-envelope alignment.
NIST	SP 800-207 Zero Trust; AI RMF 1.0; SP 800-218 SSDF. Security/AI governance anchors.
W3C	WCAG 2.2. Accessibility target.
OpenTelemetry	Observability semantic and tracing approach.
yfinance / Yahoo Finance	Current MVP stock-data adapter. Free/no-key public-unofficial access for research; upstream terms and availability apply, and no paid fallback is enabled.
Ollama	Local REST model runtime. qwen3:8b is the default review model in the companion implementation and can be changed through OLLAMA_MODEL.

Appendix D. Companion implementation quick start

The implementation ZIP is designed for copy/commit/push use. Default runtime has zero AI API charges and no paid stock-data provider. Docker still consumes local machine/cloud compute resources if you choose to host it externally.

```

cp .env.example .env
./scripts/bootstrap.sh

# Web
open http://localhost:3000

```

```
# API
open http://localhost:8000/docs

# Verify policy
curl http://localhost:8000/health
```

Runtime	Implementation
AI review	Ollama container; qwen3:8b default; local REST /api/chat
Stock data	yfinance adapter; bare IDX ticker normalized to .JK; no paid fallback
API	FastAPI
Jobs	Redis + RQ worker
Database	PostgreSQL
Web	Next.js
Operational scripts	bootstrap/start/stop/pull-model/test/smoke/backup/restore

Implementation handoff

After approval, treat this document as the v1.1 baseline synchronized to the runnable local-Ollama/free-data implementation. Changes to one-way decisions (PIT semantics, immutable published versions, canonical IDs/contracts, evidence manifest and audit format) require an Architecture Decision Record and explicit cross-functional approval. Reversible runtime choices can evolve behind the stable contracts.

End of document